

Wi-Fi Series

Data Sheet

USE_8852

Wi-Fi / BLE Module

Version

VERSION	REVISION DATE	REVISOR	REVIEWER	REVISED CONTENTS
1.0	29/07/2026	MB	AD	Initial release

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Features :

General

- Supports Wi-Fi and Bluetooth functionalities.
- Wi-Fi 6 dual-band 2T2R module.
- Bluetooth 5.4 compliant.
- PCI Express interface for WLAN and USB interface for Bluetooth.
- Enhanced Bluetooth/Wi-Fi coexistence control to improve transmission quality in different profiles.
- Integrated 32 kHz oscillator for power management.
- Dimension: 22 × 30 × 2.2 mm, typical.
- Operating temperature: 0°C to 70°C.
- Storage temperature: -55°C to 85°C.
- Supported operating systems: Android, Linux and Windows.

MCU Platform

- Chipset: RTL8852BE-CG

WLAN Features

- IEEE 802.11a/b/g/n/ac/ax compatible WLAN.
- Supports 20 MHz and 40 MHz channel bandwidth at 2.4 GHz.
- Supports 20 MHz, 40 MHz and 80 MHz channel bandwidth at 5 GHz.
- Supports IEEE 802.11ac 2×2 Wave-2 with RX MU-MIMO.
- Provides a complete IEEE 802.11n MIMO solution for both 2.4 GHz and 5 GHz bands.
- Supports IEEE 802.1X authentication methods, including EAP-TLS, TTLS, PEAP, LEAP, EAP-FAST, EAP-SIM and EAP-AKA, depending on operating-system support.
- Supports IEEE 802.11i security, including WPA, WPA2 and WPA3.
- Supports open, shared-key and pairwise-key authentication services.
- Supports S3/S4 AES/TKIP group-key updates.

Interface

- Complies with PCI Express Base Specification Revision 1.1 for WLAN.
- Supports USB 2.0 Full-Speed interface for Bluetooth.

Bluetooth Features

- Bluetooth 5.4 compliant.
- Supports Bluetooth 5 system operation.
- Supports Bluetooth 5.3 HCI Encryption Key Size Control.
- Compatible with Bluetooth v2.1 + EDR.
- Dual-mode support with simultaneous Bluetooth Low Energy and BR/EDR operation.
- Supports Class 1, Class 2 and Class 3 power-level transmission without requiring an external power amplifier.

General Description:

The USE_8852 is a highly integrated dual-band Wi-Fi 6 and Bluetooth combo module based on the Realtek RTL8852BE-CG chipset. It combines a WLAN MAC, 2T2R-capable WLAN baseband, RF circuitry and Bluetooth functionality in a single module.

The module supports two-stream IEEE 802.11ax operation with Multi-User MIMO and is compatible with IEEE 802.11a/b/g/n/ac/ax wireless LAN standards. It supports 20 MHz and 40 MHz channel bandwidths in the 2.4 GHz band and 20 MHz, 40 MHz and 80 MHz channel bandwidths in the 5 GHz band.

WLAN communication is provided through a low-power PCI Express interface compliant with PCI Express Base Specification Revision 1.1. Bluetooth communication is provided through a USB 2.0 Full-Speed interface.

The module supports Bluetooth 5.4, Bluetooth Low Energy and Bluetooth BR/EDR dual-mode operation. Enhanced Bluetooth/Wi-Fi coexistence control is included to improve transmission quality when Wi-Fi and Bluetooth are operating simultaneously.

With integrated WLAN, Bluetooth, RF and power-management functions, the USE_8852 provides a complete solution for high-performance wireless and Bluetooth applications.

For more information, contact us at info@unisemindia.com.

Pinout Diagram:

PIN	Signal	Signal	PIN
74	3_3V	GND10	15
72	3_3V	NC	13
70	NC	NC	11
68	NC	GND9	9
66	NC	NC	7
64	NC	NC	5
62	NC	GND	3
60	NC	NC	1
58	NC	NC	59
56	VL_DIS_N	GND	57
54	BT_DIS_N	PEWAKE0	55
52	PERST0	CLKREQ0	53
50	SUSCLK	GND	51
48	COEX_RXD	REFCLKN0	49
46	COEX_TXD	REFCLKP0	47
44	COEX3	GND	45
42	NC	PETN0	43
40	NC	PETP0	41
38	VENDOR DEFINED	GND	39
36	NC	PERN0	37
34	NC	PERP0	35
32	NC	GND	33
30	NC	NC	31
28	NC	NC	29
26	NC	NC	27
24	NC	NC	25
22	NC	NC	23
20	NC	NC	21
18	GND	NC	19
16	LED_2#	NC	17
14	NC	NC	15
12	NC	NC	13
10	NC	NC	11
8	NC	NC	9
6	LED_1#	GND	7
4	3_3V	USB_D-	5
2	3_3V	USB_D+	3
		GND	1

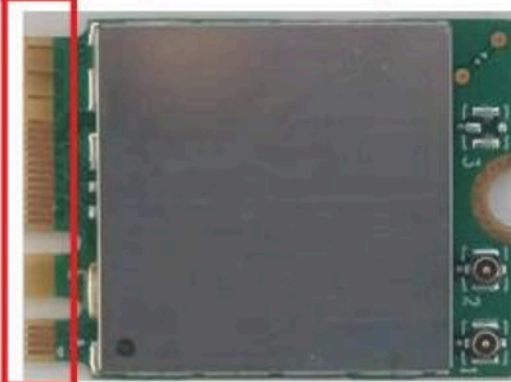


Fig 1: Module Pinouts

Pin Descriptions:

Pin No	Name	Type	Description	Voltage
1	GND	-	Ground connections	
3	USB_D+	I/O	USB differential line for BT	
5	USB_D-	I/O		
7	GND	-	Ground connections	
9	NC	-	Floating (NC)	
11	NC		Floating (NC)	
13	NC		Floating (NC)	
15	NC		Floating (NC)	
17	NC	-	Floating (NC)	

19	NC	-	Floating (NC)	
21	NC	-	Floating (NC)	
23	NC	-	Floating (NC)	
25	NC		Floating (NC)	
27	NC		Floating (NC)	
29	NC		Floating (NC)	
31	NC		Floating (NC)	
33	GND	-	Ground connections	
35	PERP0	I	PCIe RX differential signals	
37	PERN0	I		
39	GND	-	Ground connections	
41	PETP0	O	PCIe TX differential signals	
43	PETN0	O		
45	GND	-	Ground connections	
47	REFCLKP0	I	PCIe clock differential input signal	
49	REFCLKN0	I		
51	GND		Ground connections	
53	CLKREQ0	O	PCIe reference clock request signal, open drain, active low	3.3V
55	PEWAKE0	O	PCIe wake up host, open drain, active low	3.3V
57	GND	-	Ground connections	
59	NC	-	Floating (NC)	
61	NC	-	Floating (NC)	
63	GND	-	Ground connections	
65	NC	-	Floating (NC)	
67	NC	-	Floating (NC)	
69	GND9	-	Ground connections	
71	NC	-	Floating (NC)	
73	NC	-	Floating (NC)	
75	GND10	-	Ground connections	

Bottom side :

Pin No	Name	Type	Description	Voltage
2	3_3V	P	Power supply	3.3V
4	3_3V	P	Power supply	3.3V
6	LED_1#	O	WLAN LED signal	3.3V
8	NC	-	Floating (NC)	
10	NC		Floating (NC)	
12	NC		Floating (NC)	
14	NC		Floating (NC)	
16	LED_2#	O	BT LED signal	3.3V
18	GND	-	Ground connections	
20	NC	-	Floating (NC)	
22	NC	-	Floating (NC)	
24	NC		Floating (NC)	
26	NC		Floating (NC)	
28	NC		Floating (NC)	
30	NC		Floating (NC)	
32	NC		Floating (NC)	
34	NC	-	Floating (NC)	
36	NC	-	Floating (NC)	
38	VENDOR DEFINED	-	Host wake BT. No function, please don't connect to this pin.	
40	NC	-	Floating (NC)	
42	NC	-	Floating (NC)	
44	COEX3	I/O	LTE coexistence signal	3.3V
46	COEX_TXD	O	LTE coexistence signal	3.3V
48	COEX_RXD	I	LTE coexistence signal	3.3V
50	SUSCLK	I	Sleep clock input	3.3V
52	PERST0	I	PCIe reset signal, active low	3.3V
54	BT_DIS_N	I	Bluetooth enable signal, pull low to disable BT function, default high.	3.3V
56	WL_DIS_N	I	WLAN enable signal, pull low to disable BT function, default high.	3.3V

58	NC	-	Floating (NC)	
60	NC	-	Floating (NC)	
62	NC	-	Floating (NC)	
64	NC	-	Floating (NC)	
66	NC	-	Floating (NC)	
68	NC	-	Floating (NC)	
70	NC	-	Floating (NC)	
72	3_3V	P	Power supply	3.3V
74	3_3V	P	Power supply	3.3V

The signal type codes are used in the table:

- P:POWER
- I:INPUT
- O:OUTPUT

Electrical Specifications :

Power Supply DC Characteristics:

Parameter	MIN	TYP	MAX	Unit
Operating Temperature	0	25	70	deg.C
VDD33	3.0	3.3	3.6	V

Power Consumption :

Power Consumption		VDD33 = 3.3V(Unit:mA)
	TX (2.4G 11M)	369
	TX (2.4G 54M)	210
	TX (2.4G 11N)	205
	RX (2.4G 11N)	163
	TX (5G 54M)	210
	TX (5G vHT80)	187
	RX (5G vHT80)	185

RF Characteristics

1. 2.4G RF Specification

Feature	Description	
WLAN Standard	IEEE 802.11 b/g/n/ax Wi-Fi compliant	
Frequency Range	2.400 GHz ~ 2.4835 GHz (2.4 GHz ISM Band)	
Number of Channels	2.4GHz : Ch1 ~ Ch14	
Test Items	Typical Value	EVM
Output Power	802.11b /11Mbps: 19dBm $\pm$ 2 dB	EVM $\leq$ -9dB
	802.11g /54Mbps: 18dBm $\pm$ 2 dB	EVM $\leq$ -25dB
	802.11n /MCS7: 17dBm $\pm$ 2 dB	EVM $\leq$ -28dB
	802.11ax HE20 MCS11: 13dBm $\pm$ 2 dB	EVM $\leq$ -35dB
	802.11ax HE40 MCS11: 13dBm $\pm$ 2 dB	EVM $\leq$ -35dB
Spectrum Mask	Meet with IEEE standard	
Freq. Tolerance	$\pm$ 20ppm	
Test Items	TYP Test Value	Standard Value
SISO Receive Sensitivity (11b,20MHz) @8% PER	- 1Mbps @ -94 dBm	$\leq$ -83 dBm
	- 11Mbps @ -85 dBm	$\leq$ -76 dBm
SISO Receive Sensitivity (11g,20MHz) @10% PER	- 6Mbps @ -90 dBm	$\leq$ -85 dBm
	- 54Mbps @ -71 dBm	$\leq$ -68 dBm
SISO Receive Sensitivity	- MCS=0 @ -90 dBm	$\leq$ -85 dBm
(11n,20MHz) @10% PER	- MCS=7 @ -69 dBm	$\leq$ -67 dBm
SISO Receive Sensitivity (11n ,40MHz) @10% PER	- MCS=0 @ -87 dBm	$\leq$ -82 dBm
	- MCS=7 @ -66 dBm	$\leq$ -64 dBm

SISO Receive Sensitivity (11ax,20MHz) @10% PER	- MCS=0 @ -90 dBm	≤-74 dBm
	- MCS=11 @ -60 dBm	≤-52 dBm
SISO Receive Sensitivity (11ax ,40MHz) @10% PER	- MCS=0 @ -87 dBm	≤-71 dBm
	- MCS=11 @ -57 dBm	≤-49 dBm
Maximum Input Level	802.11b : -10 dBm	
	802.11g/n : -20 dBm	
Antenna Reference	Small antennas with 0~2 dBi peak gain	

2. 5GHz RF Specification

Conditions : VBAT=3.3V ; VDDIO=3.3V ; Temp:25°C

Feature	Description	
WLAN Standard	IEEE 802.11a/n/ac/ax 2x2, Wi-Fi compliant	
Frequency Range	5.150 GHz ~ 5.850 GHz (5.0 GHz ISM Band)	
Test Items	Typical Value	EVM
Output Power2	802.11a /54Mbps: 18 dBm ± 2 dB	EVM ≤ -25dB
	802.11n /MCS7: 17 dBm ± 2 dB	EVM ≤ -28dB
	802.11ac VHT20 MCS8:16 dBm ± 2 dB	EVM ≤ -30dB
	802.11ac VHT40 MCS9:15 dBm ± 2 dB	EVM ≤ -32dB
	802.11ac VHT80 MCS9:15 dBm ± 2 dB	EVM ≤ -32dB
	802.11ax HE20 MCS11:13 dBm ± 2 dB	EVM ≤ -35dB
	802.11ax HE40 MCS11:13 dBm ± 2 dB	EVM ≤ -35dB
	802.11ax HE80 MCS11:13 dBm ± 2 dB	EVM ≤ -35dB
Test Items	Test Value	Standard Value

SISO Receive Sensitivity (11a,20MHz) @10% PER	- 6Mbps @ -90 dBm	≤-85
	-54Mbps @ -71 dBm	≤-68
SISO Receive Sensitivity (11n,20MHz) @10% PER	- MCS=0 @ -90 dBm	≤-85
	- MCS=7 @ -69 dBm	≤-67
SISO Receive Sensitivity (11n,40MHz) @10% PER	- MCS=0 @ -87 dBm	≤-82
	- MCS=7 @ -66 dBm	≤-64
SISO Receive Sensitivity (11ac,20MHz)@10% PER	- MCS=0, NSS1 @ 90 dBm	≤-82
	- MCS=8, NSS1 @ -64 dBm	≤-60
SISO Receive Sensitivity (11ac,40MHz) @10% PER	- MCS=0, NSS1 @ -87 dBm	≤-79
	- MCS=9, NSS1 @ -59 dBm	≤-55
SISO Receive Sensitivity (11ac,80MHz) @10% PER	- MCS=0, NSS1 @ -84 dBm	≤-79
	- MCS=9, NSS1 @ -56 dBm	≤-54
SISO Receive Sensitivity (11ax,20MHz) @10% PER	- MCS=0 @ -90 dBm	≤-74
	- MCS=11 @ -60 dBm	≤-52
SISO Receive Sensitivity (11ax,40MHz) @10% PER	- MCS=0 @ -87 dBm	≤-71
	- MCS=11 @ -57 dBm	≤-49
SISO Receive Sensitivity (11ax,80MHz) @10% PER	- MCS=0 @ -84 dBm	≤-68
	- MCS=11 @ -54 dBm	≤-46
Maximum Input Level	802.11a/n: -30 dBm	
Antenna Reference	Small antennas with 0~2 dBi peak gain	

4. Bluetooth Specification

Feature	Description
General Specification	
Bluetooth Standard	Bluetooth V5.4 of 1, 2 and 3 Mbps.
Host Interface	USB
Antenna Reference	Small antennas with 0~2 dBi peak gain
Frequency Band	2400 MHz ~ 2483.5 MHz
Number of Channels	79 channels
Modulation	GFSK, $\pi/4$ -DQPSK, 8-DPSK

RF Specification			
	Min(dBm)	Typical(dBm)	Max(dBm)
Output Power	2	5	8
Sensitivity @ BER=0.1%for GFSK (1Mbps)			-70
Sensitivity @ BER=0.01% for $\pi/4$ -DQPSK (2Mbps)			-70
Sensitivity @ BER=0.01% for 8DPSK (3Mbps)			-70
Maximum Input Level	GFSK (1Mbps):-20dBm		
	$\pi/4$ -DQPSK (2Mbps) :-20dBm		
	8DPSK (3Mbps) :-20dBm		

Block Diagram :

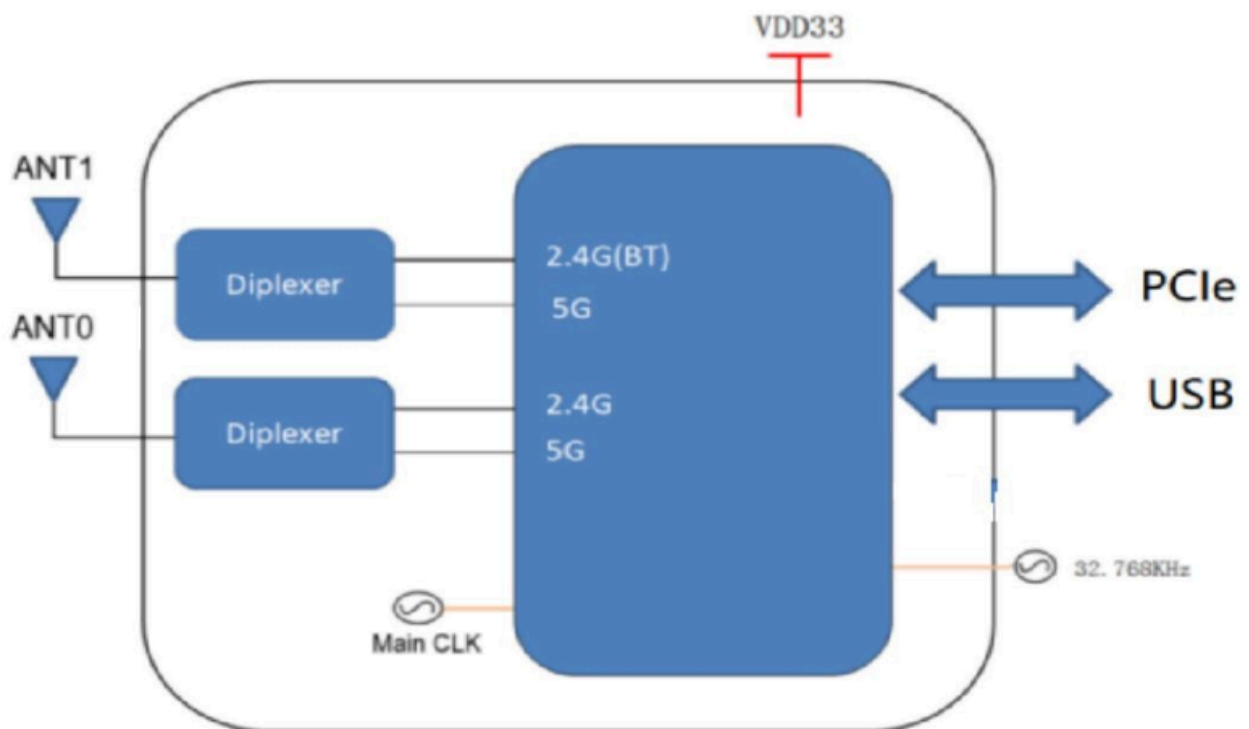


Fig 2 : Block Diagram

Reference Design :

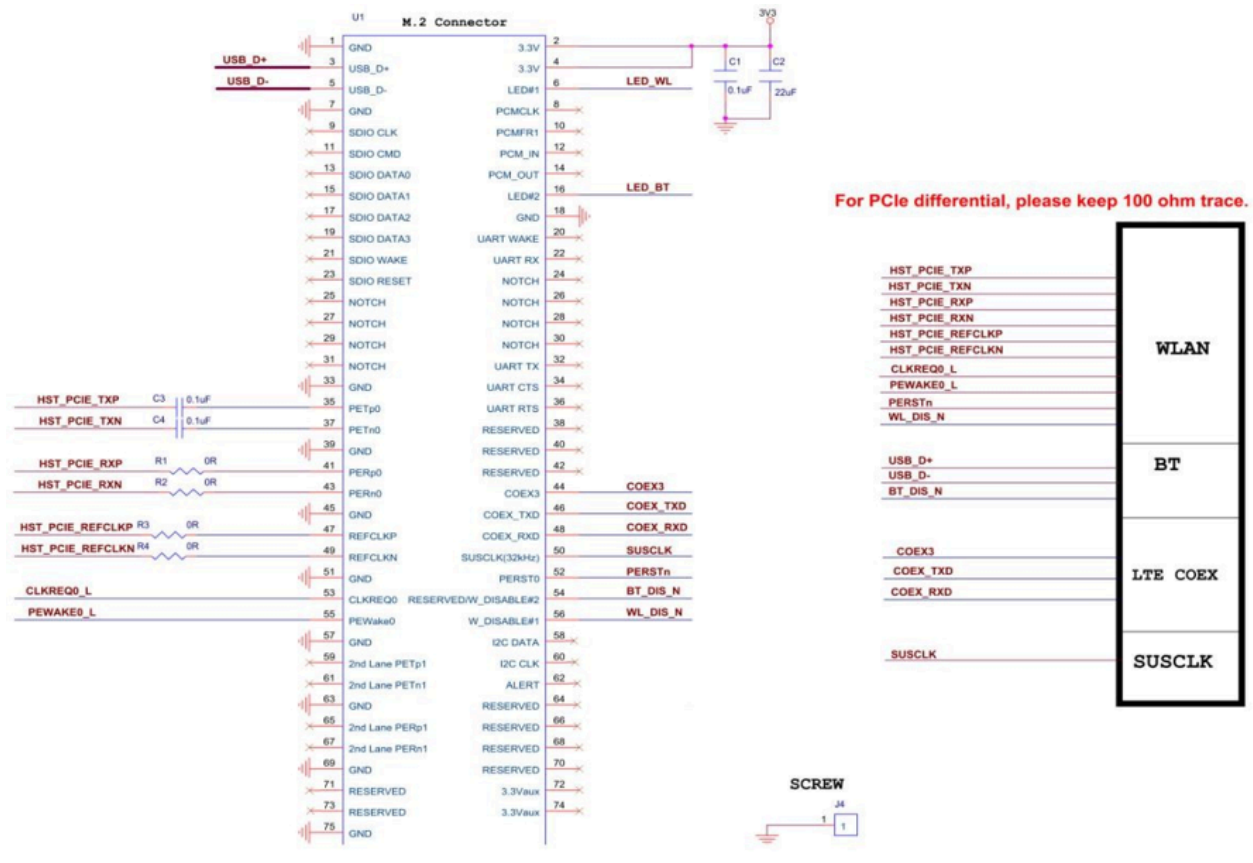


Fig 3 : Reference Design

Note:

- Both of the 2 ANT's are all support 2.4G/5G/BT function.
- The USE_8852 antenna port is controlled by the driver if the diversity function is enabled.
- C1, C2 placed close to the module side.
- The PCIe differential keeps a 100 ohm trace.
- USB differential keep 90 ohm trace

Physical Dimensions:

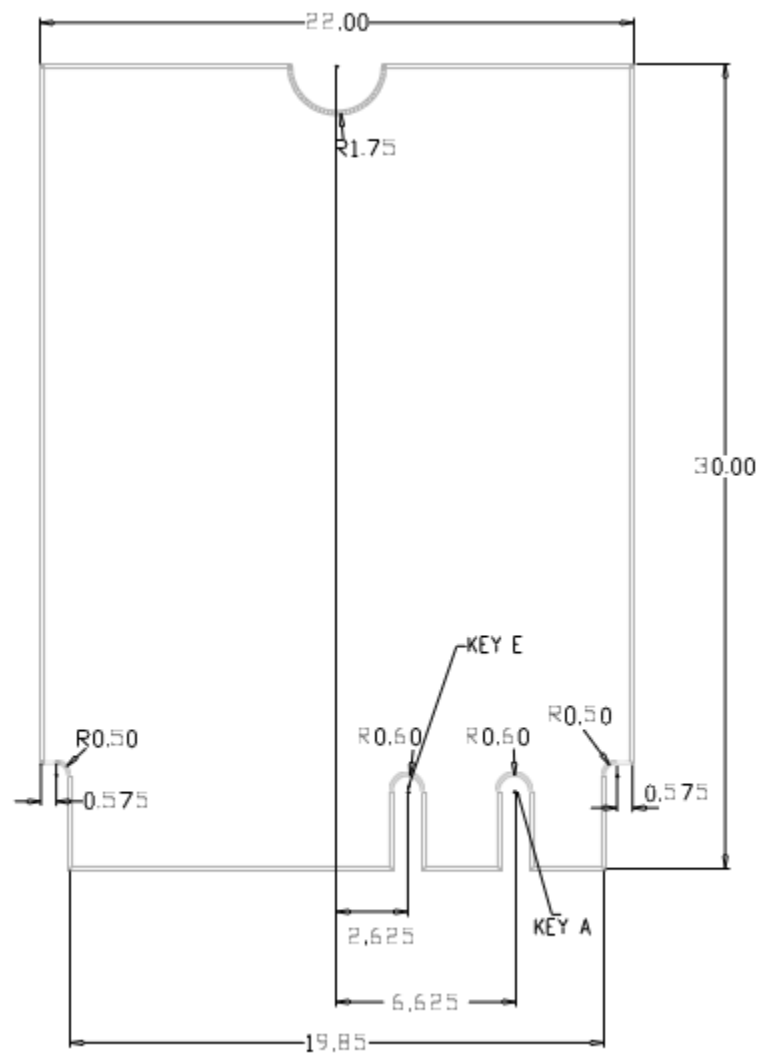


Fig 4 : Physical Dimensions

Module Dimension Drawing:

- Module Dimension:: 22 x 30 (+0.3/-0.1) mm

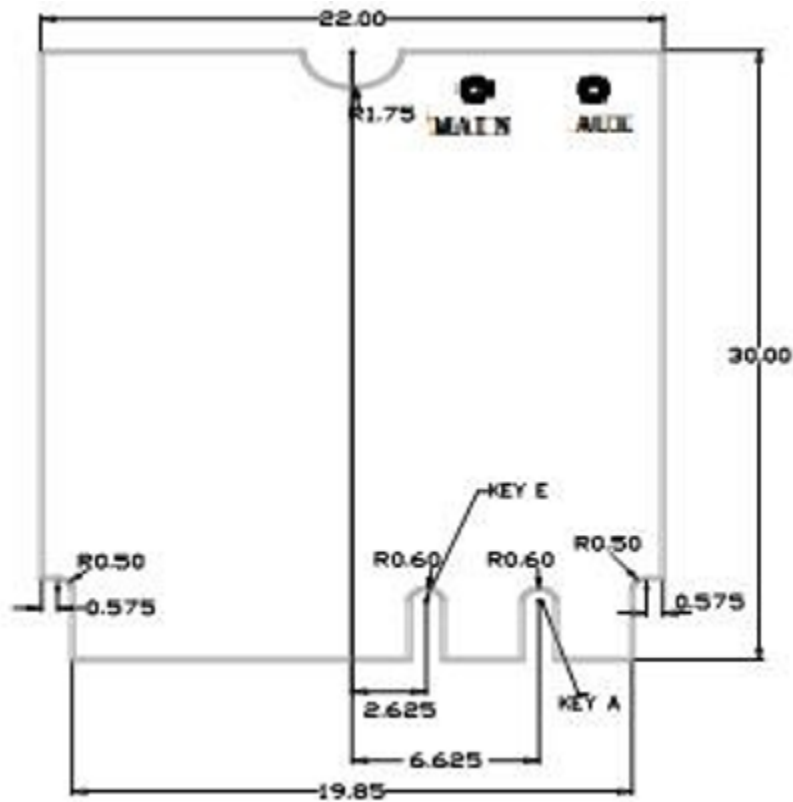


Fig 5(a) Module Dimension L × W

Product Dimension:



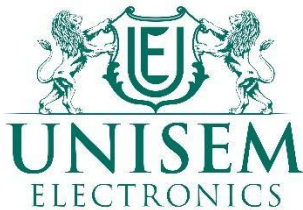
Fig 5(b) :Product Dimension

PCB thickness : 0.80mm

Product Height (Including shield) : 2.2 (±0.2) mm

Weight : 2.6g

For ordering USE_8852 please contact us on info@unisemindia.com.



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